Optipeople DemoCNC D-2800M — Process & Workflow Descriptions

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This document describes the standard processes the D-2800M is built to run, along with the typical workflow for each one. It is intended both as training material and as the reference for what the machine can realistically produce.

1. Process overview

The D-2800M is a machining center optimized for cabinet doors and drawer fronts but capable of any rectangular panel work within its envelope. Typical processes:

ID	Process	Primary spindle	Typical tools
P1	Recessed handle grooves (semicircle, rocket, horn)	S1 ATC vertical	Profile cutters Ø 8–16 mm
P2	Trunking grooves (lighting / cable channels)	S1 ATC vertical	Straight cutter Ø 4–8 mm
P3	Straightener relief grooves	S1 or S2 vertical	Straight cutter Ø 6 mm
P4	45° Lamello bevel groove	S5 bevel	Bevel cutter Ø 100 mm
P5	Lamello / biscuit holes	S6 bevel	Slot drill Ø 4 mm
P6	Hinge cup bores (top & bottom hinges)	S3 horizontal milling	Forstner-style Ø 35 mm
P7	Lock pocket and key hole	S3 horizontal milling	End mill Ø 8 mm + drill
P8	Side drilling (32 mm system)	S4 horizontal drilling	HSS drill Ø 5 / Ø 8
P9	Decorative top groove / V-cut	S1 ATC vertical	V-cutter 90° / 60°
P10	Aluminium handle bar slots	S1 ATC vertical	Carbide upcut Ø 6 mm

2. Job lifecycle

```
CAD/CAM  ──▸ OptiPanel ──▸ Job queue ──▸ Operator ──▸ Machine ──▸ Reports
  (MPR/      (import,       (barcode      (load &      (cycle)      (CMMS,
  DXF, ...)   nest, NC)      lookup)       confirm)                  ERP)
```

2.1 Where jobs come from

Three import paths are supported:

1. **MPR / MPR2 from Homag woodCAD/CAM** — most common for cabinet doors. Direct import preserves operation grouping and feeds.
2. **DXF + parameter file** — for jobs created in lighter CAD tools. The parameter file (CSV or XML) defines tool, depth, and feed per layer.
3. **OptiPanel native (.opt)** — created on the HMI or in OptiPanel Studio (desktop application).

2.2 Nesting and panel layout

For batches that don't fit a single panel, OptiPanel runs a rectangular nesting algorithm. The output is a per-panel `.opt` file with a barcode that links back to the source job.

3. Standard processes in detail

P1 — Recessed handle grooves

Cabinet doors with finger-pull handle profiles. The cutter follows a contour parallel to the top edge of the door at a depth of typically 13 mm.

Toolpath outline:

1. Plunge entry 50 mm from the door edge (avoid edge chipping).
2. Climb-mill toward the X− end of the door.
3. Lift, rapid back to start.
4. Repeat 2–3 passes to reach final depth.

Recommended parameters:

Material	Tool Ø	Feed	Spindle	DOC (per pass)
MDF 18 mm	12 mm	6 m/min	18,000 rpm	5 mm
Solid oak	12 mm	4 m/min	16,000 rpm	3 mm
Painted MDF	10 mm	3 m/min	18,000 rpm	3 mm

Quality notes:

- For painted doors, use a downcut tool to keep the painted face clean.
- Burn marks usually indicate dull cutter or insufficient feed.

P2 — Trunking grooves

A shallower, wider variant of P1. Used for built-in cable management on furniture sides.

Quality notes:

- Keep the groove ≥ 8 mm from any edge to avoid blow-out.
- For decorative furniture, dust extraction must be running well — loose chips will polish the surface inside the groove.

P4 — 45° Lamello bevel groove

Bevel cutter S5 cuts a 45° groove along an edge for Lamello T-20 connectors.

Operation sequence:

1. Beam moves to align with the workpiece edge.
2. S5 head tilts to 45° (factory-set; checked at install).
3. The cutter plunges to depth, then traverses parallel to the edge.

Common defects:

Defect	Likely cause
Uneven groove depth	Panel not properly clamped — check pressers
Burn at the start	Feed too slow at entry — adjust ramp
Width inconsistent	Cutter run-out > spec — recheck S5 collet

P6 — Hinge cup bores (top & bottom hinges)

Standard 35 mm Ø cup bores at fixed offsets from the door edges.

Specification:

- Diameter: 35 mm
- Depth: 12 mm
- Distance from top/bottom door edge: 100 mm (default; customizable per hinge brand)
- Distance from inner door edge: 22.5 mm centre

Cycle:

1. Door clamped, Z+offset auto-corrected for measured thickness.
2. S3 head positions, plunges, dwells 0.2 s, retracts.
3. Repeat for the second hinge cup.

Quality notes:

- A Ø 35 forstner bit must be sharp. Dull bits cause tear-out at the rim, especially on veneered doors.
- Hinge brand changes (Blum vs. Hettich vs. Salice) only change the offsets, not the diameter.

P7 — Lock pocket & key hole

A rectangular pocket plus a smaller bore for the key cylinder. Used on office furniture, lockable drawers, residential cabinets with locks.

Cycle:

1. End mill plunges and clears the pocket.
2. Bit change → drill.
3. Drill plunges for the cylinder hole.

P8 — Side drilling (32 mm system)

Industry-standard line drilling at 32 mm pitch, used for shelf pin holes and cabinet assembly. The S4 drilling spindle moves along the door's Y edge.

Specification:

- Pitch: 32 mm
- Diameter: 5 mm (shelf pins) or 8 mm (cam-and-dowel)
- Depth: 13 mm typical
- Edge offset: 37 mm

Cycle: one plunge per hole, ~0.4 s each. A typical 800 mm tall side panel takes 12 s for the full line of 25 holes.

4. Quality control workflow

After each cycle the operator should:

1. Visually inspect the part for obvious defects (burn marks, tear-out, missed operations).
2. Use a calibre on a representative dimension (handle groove depth, hinge cup spacing) at least once per shift, or once per setup change.
3. Tag any defective part with the job's barcode and an "NG" sticker so the rework workflow can find it.

A quarterly capability study (Cpk on hinge spacing, handle groove position) is the recommended baseline.

5. Material handling guidelines

Material	Notes
MDF	Most common. Dust is fine — extractor is mandatory
HDF	Harder, dulls tools 2× faster than MDF
Particle board	Vacuum hold is critical — porous panels need extra zones
Plywood	Reduce feed near the edges to prevent splintering
Solid wood	Grain direction matters — favour climb-milling
Lacquered MDF	Use downcut tools; cover the work zone to control overspray
Acrylic / PMMA	Reduce spindle speed and use a single-flute tool; otherwise melts
Aluminium handles	Pre-position; use carbide upcut with lubrication mist

6. Throughput planning

The machine has two work zones. The throughput-optimal workflow is:

1. Operator loads zone A.
2. Machine starts zone A. Operator immediately begins loading zone B.
3. Machine finishes zone A → automatically starts zone B.
4. Operator unloads zone A, loads next panel.

This alternating pattern keeps spindle utilization > 80 % for cycles longer than the operator's loading time (~25–35 s per panel for an experienced operator).

For very short cycles (< 30 s), the operator can't keep up and single-zone operation is acceptable.

Typical cycle times (reference)

Job profile	Cycle time
Plain MDF door, hinge cups + handle groove	38 s
Decorative door with V-grooves + handle	1 m 25 s
Drawer front with side drilling	52 s
Full kitchen door (hinges, lock, handle, decorative)	2 m 10 s
Cabinet side panel with 32mm pin line	1 m 05 s

Above figures assume MDF 18 mm, sharp tools, and warm machine.

7. Common workflows

7.1 First piece of a new batch

1. Operator scans the first panel's barcode.
2. HMI shows job preview. Operator confirms tool list matches the magazine (HMI flags any mismatch with a red dot).
3. Cycle runs. Operator watches it through.
4. Inspect the finished part. Adjust any per-job offsets if needed (e.g. hinge position) in **Jobs** → **Edit** → **Per-batch offsets**.
5. Saved offsets apply automatically to the rest of the batch.

7.2 Tool wear mid-batch

1. HMI raises `W-4001 Tool life expired` or operator notices burn marks.
2. Pause at end of current cycle (do not abort mid-cycle).
3. Change tool. Re-calibrate length (operator manual §6.2).
4. Reset the life counter.
5. Resume.

7.3 Panel reject and rerun

If a part comes off defective:

1. Mark it NG and remove from the line.
2. From **Jobs** → **History**, find the cycle, click **Rerun**.
3. The same NC is queued — load a fresh panel and proceed.
4. The original cycle is kept in history with a "rerun" link, so the shop floor can see why an extra panel was made.

7.4 End of day / start of day handover

The HMI's **End of Shift** report summarizes:

- Cycles completed
- Tool changes
- Alarms (warnings + errors + faults)
- Spindle hours added to each spindle's TBO counter
- Top 5 jobs by cycle count

This report can be auto-emailed to the production manager — configure under **Settings** → **Reports** → **Shift Email**.

8. Integration scenarios

8.1 With Homag woodCAD/CAM

MPR/MPR2 export drops a file into a watched folder. OptiPanel imports, generates the NC, and queues the job by barcode. Operator scans the printed sticker on the panel to start.

8.2 With an ERP / MES

The machine exposes a Modbus TCP map (optional OP-OPT-IO module) with current cycle, current job ID, total cycles today, and alarm state. Some shops pull these once per minute into Power BI or a similar dashboard.

8.3 With Opti.AI Chat / Optipeople platform

The machine is registered as "CNC Drilling" in the Optipeople admin panel. Operators and supervisors can ask natural-language questions ("what does E-3210 mean?", "how often should I grease the X axis?") and the assistant resolves answers against the documents in this knowledge base.